

Three-Point Attempt Rate and Winning Outcomes: Evidence from the NBA (2005–2024)

Haotian Chang

September 2026

Abstract

Does shooting more three-pointers win NBA games, and has that return changed as the whole league embraced the three? Using a panel of 600 team-seasons from 2005 to 2024 with team and season fixed effects, this paper finds that the answer depends on a single modeling choice. The original model, which controlled for offensive and defensive efficiency, found no effect at all. Yet those measures are outcomes of the same season's play that sit between shot selection and winning, so holding them fixed blocks the very channel through which three-point shooting affects winning outcomes. A new model without these controls shows that each percentage point increase in a team's three-point attempt rate is associated with 0.8 more wins per season ($p < 0.001$). The return was largest in the mid-2000s and appears to erode as the rest of the league shoots more threes, although the decline is not statistically significant. The advantage of three-point shooting, in other words, was available long before most teams acted on it.

Keywords: sports economics; technology diffusion; panel data; fixed effects; bad controls

1. Introduction

Over the past two decades, the National Basketball Association (NBA) has undergone a profound structural shift in offensive strategy, moving from traditional interior scoring to a perimeter-oriented “small-ball” approach that prioritizes spacing, transition speed, and three-point volume over interior height and post-up play. The shift is captured most directly by the three-point attempt rate (3PA%), three-point attempts as a proportion of all field-goal attempts, which roughly doubled over this period. It also offers a clean setting for studying how a strategic innovation diffuses and whether it eventually saturates. This paper asks how the marginal effect of 3PA% on team winning outcomes has evolved across the 2005 to 2024 NBA seasons. The hypothesis is that the effect is nonlinear: increasing returns as early adopters optimized the strategy, followed by decline as the strategy saturated league-wide.

This research originated as a collaborative project in *ECON 4831: Sports Data Analytics and Economic Analysis*; the current version has been expanded and refined independently by the author. I construct a balanced panel of 600 team-season observations and estimate a quadratic-interaction model by ordinary least squares:

$$\begin{aligned} WinRate_{it} = & \alpha_i + \gamma_t + \beta_1 \cdot 3PA\%_{it} + \beta_2 \cdot 3PA\%_{it}^2 \\ & + \beta_3 \cdot (3PA\%_{it} \times YearIndex_t) + \beta_4 \cdot (3PA\%_{it} \times YearIndex_t^2) + \varepsilon_{it} \end{aligned}$$

This model enables the marginal effect of 3PA% to vary over time, capturing the rise and saturation of the small-ball era. It includes team and season fixed effects to isolate the strategic impact of shot selection and control for time-invariant team characteristics and broader league-wide trends; standard errors are clustered by team. Offensive and defensive efficiency are deliberately excluded, since both are realized outcomes of the same season’s play and lie on the causal pathway from three-point shooting to winning. The marginal effect is computed as the first derivative of win rate with respect to 3PA%, evaluated season by season.

Three findings follow. First, the return to three-point volume is positive and robust: each percentage increase in 3PA% is associated with 0.97 percentage increase in win rate, roughly 0.8 wins per season. Second, this return diffuses in specifications that account for efficiency measures, an issue of bad control

that generalizes beyond basketball. Third, the estimated effect declines toward zero by the 2020s, consistent with the hypothesized saturation of small-ball, but the tests cannot reject a constant effect and no period of increasing returns appears anywhere in the sample. Similar to the “Moneyball” era in baseball, the small-ball revolution reflects a market-wide correction in the marginal productivity of an undervalued input, where the evidence points to adjustment costs, not information, as the binding constraint on diffusion.

2. Data

The foundation of this study is a balanced panel of 600 team-season observations. Data were retrieved from [Basketball-Reference](#) and cover the 2005 through 2024 NBA regular seasons. This period was selected to capture the full trajectory of the modern offensive transformation: it begins in the pre-analytics era of the mid-2000s, through the rise of the small-ball era in the mid-2010s, and ends in the contemporary period of high-volume three-point shooting.

The raw data were cleaned in two steps. First, franchise identity was standardized across relocations and renaming, so that the Seattle SuperSonics and Oklahoma City Thunder, the New Jersey Nets and Brooklyn Nets, the Charlotte Bobcats and Charlotte Hornets, and the New Orleans Hornets and New Orleans Pelicans are each treated as one continuous entity. Without this step the data contain 35 nominal team names rather than 30 franchises, and the team fixed effects would treat a relocation as the exit of one organization and the entry of another. After standardization, every team is observed in all twenty seasons and the panel is balanced. Second, all variables are expressed as rates instead of counts, so that the shortened 2012 lockout season (66 games) and the shortened 2020 and 2021 pandemic-affected seasons (64 to 75 games and 72 games respectively) remain comparable to full 82-game seasons. Data are aggregated at the team-season level in order to study strategic outcomes and league-wide trends rather than individual player volatility. The notation and definition of each variable are summarized in Table 1.

Table 1. Description of Variables

Variable	Description
<i>Indices and dependent variable</i>	
i	Team index. Thirty teams with identity held fixed across relocation and renaming.
t	Season index. The twenty regular seasons from 2005 through 2024, each labeled by the calendar year in which it ends.
WinRate_{it}	Winning percentage: games won divided by games played by team i in season t .
<i>Independent variables</i>	
$3\text{PA}\%_{it}$	Three-point attempt rate: three-point attempts divided by total field-goal attempts ($3\text{PA}/\text{FGA}$).
$3\text{PA}\%^2_{it}$	Square of the attempt rate. Included to test for curvature in the return to three-point volume.
YearIndex_t	Season counter, equal to 1 in 2005 and 20 in 2024. Entered in estimation centered at the sample midpoint, 10.5.
$3\text{PA}\% \times \text{YearIndex}$	Interaction of attempt rate with the season counter. Allows the marginal effect of $3\text{PA}\%$ to shift linearly over the sample.
$3\text{PA}\% \times \text{YearIndex}^2$	Interaction with the squared season counter. Allows the marginal effect to rise and then fall, as the saturation hypothesis predicts.
<i>Fixed effects and error term</i>	
α_i	Team fixed effect. Absorbs time-invariant organizational characteristics such as market size, ownership, and persistent front-office philosophy.
γ_t	Season fixed effect. Absorbs league-wide shocks common to all teams in a season, including rule and officiating changes and the secular rise in three-point volume.
ε_{it}	Idiosyncratic error. Standard errors are clustered at the franchise level.
<i>Post-treatment variables</i>	
ORtg_{it}	Offensive Rating: points scored per 100 possessions.
DRtg_{it}	Defensive Rating: points allowed per 100 possessions.
Pace_{it}	Possessions per 48 minutes.

Notes: Offensive Rating, Defensive Rating, and Pace are used in one regression. Each is realized in the same season's play and lies between three-point shooting and winning, which blocks the channel through which $3\text{PA}\%$ operates.

Panel A of Table 2 captures the strategical shift over the sample period. The three-point attempt rate varies widely, from a minimum of 0.105 (Los Angeles Clippers, 2005) to a maximum of 0.519 (Houston Rockets, 2019), with a sample mean of 0.288. Winning percentage is centered on 0.500 by construction, since league-wide wins and losses must balance. Offensive rating rises substantially over the period, reaching a maximum of 123.2 points per 100 possessions (Boston Celtics, 2024).

Panel B describes two features of this variation. First, most of the variation in three-point attempt rate is a league-wide time trend rather than differences between teams within a season: 74.6 percent of its total variance lies between seasons, and once team and season fixed effects are removed only about a fifth of the original variance remains, with a residual standard deviation of 0.038. The estimates in Section 4 are therefore identified from a team's deviation from both its own long-run three-point attempt rate and the

average level of the league within a season, not from the rise in three-point volume which the season fixed effects absorb in full.

Second, teams have converged on a common three-point attempt rate. The coefficient of variation of 3PA% within a season falls from 0.256 in 2005 to 0.072 in 2024, a decline of roughly 0.008 per season ($p < 0.001$). In absolute terms the within-season standard deviation shows no statistically significant trend over the period (0.050 in 2005 against 0.028 in 2024, trend $p = 0.13$), so the convergence is relative rather than absolute: the spread across teams has narrowed only modestly while the average level of the league has doubled. This pattern is consistent with the saturation hypothesis. It also implies that progressively less cross-sectional variation is available to identify the effect of three-point shooting in the most recent seasons, which is reflected in the widening confidence intervals reported in Table 4.

Table 2. Summary Statistics and Sources of Variation (2005–2024)

<i>Panel A. Summary statistics</i>					
Variable	Obs.	Mean	Std. Dev.	Min	Max
WinRate	600	0.500	0.150	0.106 (CHA, 2012)	0.890 (GSW, 2016)
3PA%	600	0.288	0.085	0.105 (LAC, 2005)	0.519 (HOU, 2019)
ORtg	600	108.56	4.39	95.20 (CHA, 2012)	123.20 (BOS, 2024)
DRtg	600	108.56	4.19	98.20 (BOS, 2012)	120.40 (UTA, 2024)
Pace	600	94.91	3.99	86.30 (MEM, 2006)	105.10 (MIL, 2020)
<i>Panel B. Sources of variation</i>					
Variable	Total SD	Share between seasons	SD after two-way FE	Variance absorbed by FE	
WinRate	0.150	0.0%	0.138	15.9%	
3PA%	0.085	74.6%	0.038	80.3%	
ORtg	4.390	46.2%	2.937	55.2%	
DRtg	4.188	50.7%	2.649	60.0%	
Pace	3.995	69.6%	1.865	78.2%	

Notes: Each observation is one franchise-season from 2005 to 2024; $N = 600$ throughout. WinRate and 3PA% are expressed as proportions. ORtg is points scored per 100 possessions and DRtg is points allowed per 100 possessions, so lower values of DRtg indicate better defense. Pace is possessions per 48 minutes.

Panel B decomposes the variation in each variable. "Between seasons" is the share of total variance accounted for by season means, that is, by the league-wide time path. "SD after 2-way FE" is the standard deviation of the residual from a regression of the variable on franchise and season indicators, and is the variation actually used for identification. "Variance absorbed" is the share of total variance removed by the two sets of fixed effects. WinRate has a between-season share of zero because the league mean is 0.500 in every season by construction.

3. Methodology

3.1 Specification

The research question asks how the return to three-point volume has evolved over the same period, so the estimating equation must allow the marginal effect of the attempt rate to change over time. Therefore, I estimate a quadratic-interaction panel model. The squared three-point attempt rate allows the return to volume to diminish as volume rises, which is the form that a saturation point would take within a single season. The interactions of the attempt rate with a linear and a quadratic season index allow the entire return schedule to shift across seasons, which is the form that saturation would take across the sample. The coefficients on these two terms are unrestricted in sign, so how the effect evolves is estimated rather than assumed. The specification is

$$\begin{aligned} WinRate_{it} = & \alpha_i + \gamma_t + \beta_1 \cdot 3PA\%_{it} + \beta_2 \cdot 3PA\%^2_{it} \\ & + \beta_3 \cdot (3PA\%_{it} \times YearIndex_t) + \beta_4 \cdot (3PA\%_{it} \times YearIndex^2_t) + \varepsilon_{it} \end{aligned}$$

where i denotes the 30 teams and t the 20 seasons from 2005 to 2024. $YearIndex$ runs from 1 to 20 and is centered at the sample midpoint of 10.5 before the interactions are formed. Here, centering is a reparameterization that does not change the fitted values, marginal effects, and test statistics. It substantially reduces the correlation among the interaction terms and makes the coefficient on three-point attempt rate interpretable at the middle of the sample rather than at an unobserved season zero. The model is estimated by ordinary least squares with a full set of team and season indicators.

Team fixed effects control for time-invariant characteristics such as market size and managerial strategy used by the teams. Without this control, the estimates could reflect the possibility that successful organizations simply shoot more three-pointers. Season fixed effects control for league-wide shocks, including the league-wide rise in three-point volume itself, which accounts for about three quarters of the total variation in 3PA%. As Panel B of Table 2 shows, the season effects alone absorb roughly three quarters of the variance in the attempt rate. What identifies the coefficients is the residual variation: a franchise's deviation from both its own long-run three-point attempt rate and the average level of its season. This is the

appropriate comparison for the question at hand, which asks whether shooting more three-pointers than one's contemporaries is rewarded, not whether the league as a whole has shot more of them over time.

In addition to the full specification, I also report a restrictive version that retains only the linear term in the attempt rate. The full specification answers the question of how the return has evolved; the linear version summarizes the average return over the sample in a single interpretable coefficient, and serves as the reference point against which the more flexible models are judged.

3.2 The exclusion of same-season performance measures

Offensive Rating, Defensive Rating, and Pace are available for every team-season and are reported in Table 2 but they are excluded from the estimating equation: they are not pre-determined controls. Both are realized outcomes of the same season's play and lie on the causal path running from shot selection to winning. A team that shifts its shot distribution toward three-pointers changes its points scored per possession, which, in terms of offensive efficiency, is precisely the mechanism through which the strategic choice affects results.

Controlling for a variable that lies between the treatment and the outcome is a classic bad-control problem. The resulting coefficient no longer answers the research question; rather, it estimates how much a team's win rate changes when it shoots more three-pointers while its offensive and defensive efficiency are held constant, and the answer to that question is close to nothing since the channel through which the strategy operates has been shut off by construction. The effect is substantial and is reported in Table 3: the coefficient on the attempt rate is 0.968 with a standard error of 0.237 when offensive and defensive efficiency are omitted, and falls to statistically insignificant value of -0.061 once they are included. Given these numbers, drawing the conclusion that three-point shot volume does not matter would be to mistake a modeling choice for an empirical finding.

Pace is excluded for the same reason. A team's number of possessions per 48 minutes is a strategy made together with its shot distribution instead of a fixed characteristics of its environment, so controlling for it absorbs part of the same decision the model is trying to measure. A specification including all three

measures is reported in the final column of Table 3 as a sensitivity check and to document the magnitude of the problem, but no causal inference is drawn from it.

3.3 Inference

Observations within a team are unlikely to be independent across seasons. Rosters, coaching staff, and managerial strategies persist, so the unexplained component of a team's performance in one season is correlated with the next. Conventional heteroskedasticity-robust standard errors therefore do not address this dependency and will generally be too small. To fix that problem, I cluster the standard errors at the team level, which allows for arbitrary correlation of the errors within a team over the full twenty seasons while maintaining independence across different teams. Because the team fixed effects are nested within the clustering variable, they are treated as redundant in the degrees-of-freedom adjustment, and causal inference are drawn based on the t-distribution with 29 degrees of freedom.

This model choice is conservative: the clustered standard errors are roughly 1.6 times the heteroskedasticity-robust ones, so the significance levels reported below are lower than they would otherwise be. With 30 clusters the asymptotic justification is sufficient but not generous, and results that are marginal under clustering should be read with corresponding caution. Heteroskedasticity-robust standard errors are reported as a sensitivity check and do not change the sign or which coefficients are statistically significant.

3.4 Marginal effects and hypothesis tests

Since the three-point attempt rate enters the equation four times, the research question is answered not by any single coefficient but by the marginal effect: the first derivative of the win rate with respect to the attempt rate, depending on both the level of the attempt rate and the season:

$$ME(3PA\%, t) = \frac{\partial WinRate_{it}}{\partial 3PA\%_{it}} = \beta_1 + 2\beta_2 \cdot 3PA\%_{it} + \beta_3 \cdot YearIndex_t + \beta_4 \cdot YearIndex_t^2$$

I evaluate this expression at each season's league-mean attempt rate and obtain standard errors by the delta method from the clustered variance matrix, with confidence intervals based on the t distribution with 29 degrees of freedom. Reporting the derivative rather than the individual coefficients is not merely a matter of presentation. After the fixed effects are partialled out, the correlation between the attempt rate and its square is 0.96 and their variance inflation factors are 61 and 66, because the attempt rate is observed only over the interval from 0.105 to 0.519 and is close to linear in its own square on that interval. Their coefficients are therefore imprecisely estimated and should not be interpreted separately. The centered interaction terms, by contrast, are only modestly collinear, with variance inflation factors of 5.5 and 2.4. The derivative, being a linear combination of all four coefficients, is estimated far more precisely, and the joint tests below are informative where the individual t-statistics are not.

Three hypotheses are tested on the full specification. The first is that the attempt rate has no effect on winning in any form, which requires all four coefficients to be zero jointly and is tested by an F test rather than by inspection of individual t-statistics. The second is the saturation hypothesis proper, that the marginal effect does not vary over the sample period, which requires the null that the third and fourth coefficients are jointly zero and is likewise an F test; the original formulation of this test as a pair of separate t-tests is not valid, since the two terms are jointly determined and highly correlated. The third is that the return schedule is linear in the attempt rate within a season, which tests the second coefficient alone.

Finally, the parametric specification imposes a particular shape on the time path: the marginal effect is constrained to be a quadratic function of the season counter. To check whether the results depend on that restriction, I also estimate an unrestricted alternative in which the attempt rate is interacted with a full set of season indicators,

$$WinRate_{it} = \alpha_i + \gamma_t + \sum_s \theta_s \cdot (3PA\%_{it} \times 1[t = s]) + \varepsilon_{it}$$

so that each season receives its own slope and no functional form is imposed on how the return evolves. This specification is less efficient, since it estimates twenty parameters where the parametric model estimates four, but it provides a direct check on whether any apparent time path is a feature of the data or

an artifact of the quadratic restriction. The two sets of estimates are reported together in Table 4 and plotted in Figure 1.

4. Results

This section analyzes the results of the quadratic and interaction models, tracing how the relationship between perimeter shot volume and competitive success has shifted as the league approached strategic maturity. The analysis is presented across five specifications in Table 3, moving from a simple cross-sectional correlation to a fixed-effects interaction framework. Columns (1) through (3) establish the average return to three-point volume and test for curvature within a season. Column (4) is the preferred specification and is central to the research question, since it allows the marginal effect of the three-point attempt rate to vary dynamically across the twenty-season sample. Column (5) reproduces the earlier practice of conditioning on contemporaneous efficiency measures and is reported only to document the consequences of that choice. The season-by-season marginal effects implied by column (4) are reported in Table 4 and plotted in Figure 1.

4.1 The Average Return to Three-Point Volume

- **Baseline (Column 1):** in the pooled quadratic model, neither the linear term (0.892, standard error 0.626) nor the quadratic term (-1.079, standard error 1.054) is statistically distinguishable from zero, and the model explains only 2.3 percent of the variation in win rate. This is the expected result rather than a disappointing one. The pooled comparison places a low-volume team from 2005 alongside a high-volume team from 2024 and treats the difference between them as informative, when in fact the two are separated by two decades of roster composition, rule enforcement, and league-wide strategy.

Table 3. Regression Results: Three-Point Attempt Rate and Win Rate

	(1)	(2)	(3)	(4)	(5)
	<i>Quadratic</i>	<i>Linear + FE</i>	<i>Quadratic + FE</i>	<i>Full + FE</i>	<i>Full + FE + controls</i>
	Win Rate	Win Rate	Win Rate	Win Rate	Win Rate
3PA%	0.892 (0.626)	0.968*** (0.237)	1.336* (0.615)	-1.015 (1.762)	-0.393 (0.427)
3PA% ²	-1.079 (1.054)		-0.639 (1.012)	3.706 (2.874)	0.556 (0.742)
3PA% x YearIndex				-0.144 (0.090)	-0.022 (0.018)
3PA% x YearIndex ²				-0.0075 (0.0058)	-0.0000 (0.0016)
Offensive Rating					0.0301*** (0.0007)
Defensive Rating					-0.0297*** (0.0008)
Pace					-0.0018* (0.0008)
Team Fixed Effects	No	Yes	Yes	Yes	Yes
Season Fixed Effects	No	Yes	Yes	Yes	Yes
Performance controls	No	No	No	No	Yes
Observations	600	600	600	600	600
R ²	0.023	0.218	0.219	0.228	0.943
Within R ²	0.023	0.070	0.071	0.082	0.932

Cluster-robust standard errors, clustered by franchise (30 clusters), in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001.

Notes: The panel contains 600 franchise-season observations covering 30 franchises over the 2005-2024 regular seasons. YearIndex is centered at 10.5. Column (4) is the preferred specification. Its four strategy terms are jointly significant (F = 8.35, p = 0.0001) although none is individually significant; the null of no time variation cannot be rejected (F = 1.62, p = 0.215), nor the null of no curvature (F = 1.66, p = 0.207). Column (5) conditions on post-treatment performance measures and is reported for comparison only.

Table 4. Marginal Effect of Three-Point Attempt Rate on Win Rate, by Season

Season	Mean 3PA%	<i>Quadratic-interaction model, col. (4)</i>			<i>Unrestricted season slope</i>	
		ME	SE	95% CI	Slope	SE
2005	0.196	0.0113**	(0.0040)	[0.0032, 0.0194]	0.0106*	(0.0052)
2006	0.202	0.0117**	(0.0032)	[0.0050, 0.0183]	0.0066	(0.0047)
2007	0.212	0.0122***	(0.0027)	[0.0066, 0.0177]	0.0099*	(0.0044)
2008	0.222	0.0125***	(0.0024)	[0.0075, 0.0175]	0.0166*	(0.0067)
2009	0.224	0.0121***	(0.0024)	[0.0072, 0.0170]	0.0177**	(0.0064)
2010	0.222	0.0113***	(0.0027)	[0.0057, 0.0168]	0.0136*	(0.0063)
2011	0.222	0.0104**	(0.0032)	[0.0038, 0.0170]	0.0124*	(0.0051)
2012	0.226	0.0097*	(0.0037)	[0.0023, 0.0172]	0.0073	(0.0051)
2013	0.243	0.0099*	(0.0036)	[0.0025, 0.0173]	0.0146*	(0.0066)
2014	0.260	0.0098*	(0.0036)	[0.0024, 0.0172]	0.0117	(0.0073)
2015	0.268	0.0090*	(0.0039)	[0.0011, 0.0169]	0.0121	(0.0065)
2016	0.284	0.0086*	(0.0038)	[0.0008, 0.0164]	0.0087	(0.0062)
2017	0.316	0.0092**	(0.0033)	[0.0024, 0.0160]	0.0069	(0.0052)
2018	0.337	0.0089**	(0.0032)	[0.0023, 0.0155]	0.0061	(0.0076)
2019	0.359	0.0084*	(0.0032)	[0.0020, 0.0149]	0.0101	(0.0051)
2020	0.384	0.0081*	(0.0032)	[0.0015, 0.0147]	0.0087	(0.0057)
2021	0.392	0.0064	(0.0036)	[-0.0011, 0.0138]	0.0025	(0.0066)
2022	0.400	0.0045	(0.0043)	[-0.0043, 0.0132]	-0.0016	(0.0079)
2023	0.388	0.0010	(0.0057)	[-0.0107, 0.0127]	0.0070	(0.0058)
2024	0.395	-0.0013	(0.0068)	[-0.0153, 0.0126]	0.0096	(0.0098)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All entries give the change in win rate associated with a one percentage point increase in 3PA%. Over an 82-game season, 0.0100 corresponds to approximately 0.82 additional wins.

Notes: Columns 3-5 report the marginal effect implied by column (4) of Table 3, evaluated at each season's league-mean attempt rate, with delta-method standard errors from the clustered variance matrix; inference uses the t distribution with 29 degrees of freedom, matching the number of clusters less one. Columns 6-7 report an unrestricted alternative in which 3PA% is interacted with a full set of season indicators. The twenty unrestricted slopes are jointly different from zero ($F = 2.53$, $p = 0.011$) but cannot be rejected as all equal ($F = 0.97$, $p = 0.514$).

The direction of this shift deserves emphasis, because it is the opposite of what a spurious-correlation account would predict. If the pooled association had been driven by the possibility that well-run organizations happen both to win and to shoot more three-pointers, absorbing franchise identity would have weakened it. Instead the estimate strengthens and sharpens. The pooled comparison was attenuated rather than inflated, because it mixed variation across eras and across organizations with the within-franchise, within-season variation that actually speaks to the strategic question. Shooting more three-pointers than one's contemporaries is rewarded; the effect was simply obscured by the wrong comparison.

- **Curvature within a season (Column 3):** adding the squared attempt rate leaves the linear term positive and significant (1.336, standard error 0.615) and yields a negative but imprecise quadratic

term (-0.639, standard error 1.012). The null of no curvature cannot be rejected ($F = 1.66$, $p = 0.207$). Within the range of attempt rates actually observed, from 0.105 to 0.519, the data provide no evidence of an interior saturation point at which additional three-point volume begins to reduce winning. The return schedule appears approximately linear over the observed support, which does not rule out diminishing returns beyond it.

- **Contemporaneous efficiency measures (Column 5):** when Offensive Rating, Defensive Rating, and Pace are included, the coefficient on three-point attempt rate collapses from 0.968 to -0.061 and loses all significance, while the two efficiency measures themselves take coefficients of 0.030 and -0.030 and are significant beyond the 0.1 percent level. As argued in Section 3.2, this collapse is a mechanical consequence of conditioning on the mechanism rather than evidence about shot selection. Offensive and defensive rating are close to a restatement of the outcome: a regression of win rate on net rating alone explains 93.5 percent of the variance in the sample. Holding them fixed leaves almost nothing for any strategic variable to explain.

4.2 Time Variation in the Marginal Effect

Column (4) estimates the full specification, in which the attempt rate, its square, and its interactions with the linear and quadratic season counter appear together. This is the first specification in which all four strategy terms are estimated jointly, and the four are jointly significant ($F = 8.35$, $p = 0.0001$). No individual coefficient reaches conventional significance. For the attempt rate and its square this reflects the collinearity documented in Section 3.4; for the interaction terms, which are only modestly collinear, it reflects genuinely weak evidence of time variation. The derivative reported in Table 4 is estimated far more precisely than any of its components.

The central hypothesis of the paper concerns the interaction terms. The null that the marginal effect does not vary over the sample period cannot be rejected ($H_0: b_3 = b_4 = 0$; $F = 1.62$, $p = 0.215$). Subject to that important qualification, the estimated path of the marginal effect is as follows:

- **Mid-2000s:** contrary to the initial hypothesis, the return to three-point volume is at its highest in the earliest seasons of the sample. The marginal effect is 0.0113 in 2005 and peaks at 0.0125 in 2008, significant at the 1 percent level or better in every season of the late 2000s. Shooting more three-pointers than one's contemporaries was rewarded before the small-ball era began, not only after it.
- **Mid-2010s:** the marginal effect declines gradually but remains positive and statistically significant throughout, falling from 0.0104 in 2011 to 0.0090 in 2015 and 0.0086 in 2016. There is no period of increasing returns corresponding to the competitive peak of the small-ball revolution. The early adopters of this period were not exploiting an advantage that had newly appeared; they were the first to act on an advantage that the estimates indicate had been available throughout.
- **Early 2020s:** the marginal effect continues to decline and ceases to be statistically distinguishable from zero after 2020, falling from 0.0081 in 2020 to 0.0064 in 2021, 0.0045 in 2022, and approximately zero in 2023 and 2024. This is the pattern the saturation hypothesis predicts, and it is reinforced by the descriptive convergence documented in Section 2: the coefficient of variation of the attempt rate within a season falls from 0.256 in 2005 to 0.072 in 2024, so that by the end of the sample teams differ from one another in shot distribution far less than they once did.

Two considerations restrain the interpretation of this path. The first is the formal test already noted: the estimated decline, although visible in Figure 1, is not statistically distinguishable from a constant marginal effect. The second is a check on functional form. Estimating a separate slope for every season, which imposes no shape on the time path, yields twenty coefficients that are jointly different from zero ($F = 2.53$, $p = 0.011$) but that cannot be rejected as being all equal to one another ($F = 0.97$, $p = 0.514$). The unrestricted estimates, plotted alongside the parametric path in Figure 1, are individually imprecise and do not trace the smooth decline that the quadratic specification produces. Part of this imprecision is itself a consequence of convergence: as teams became more alike in attempt rate, less cross-sectional variation remained with which to identify the effect in recent seasons, and the confidence intervals in Table 4 widen accordingly.

The results are therefore best summarized as a robust average finding accompanied by a suggestive but unconfirmed dynamic one. Three-point attempt rate has a positive and statistically significant association with winning across the sample as a whole. The point estimates decline monotonically over the period and reach zero in the most recent seasons, which is consistent with saturation, but the sample does not permit that decline to be established statistically.

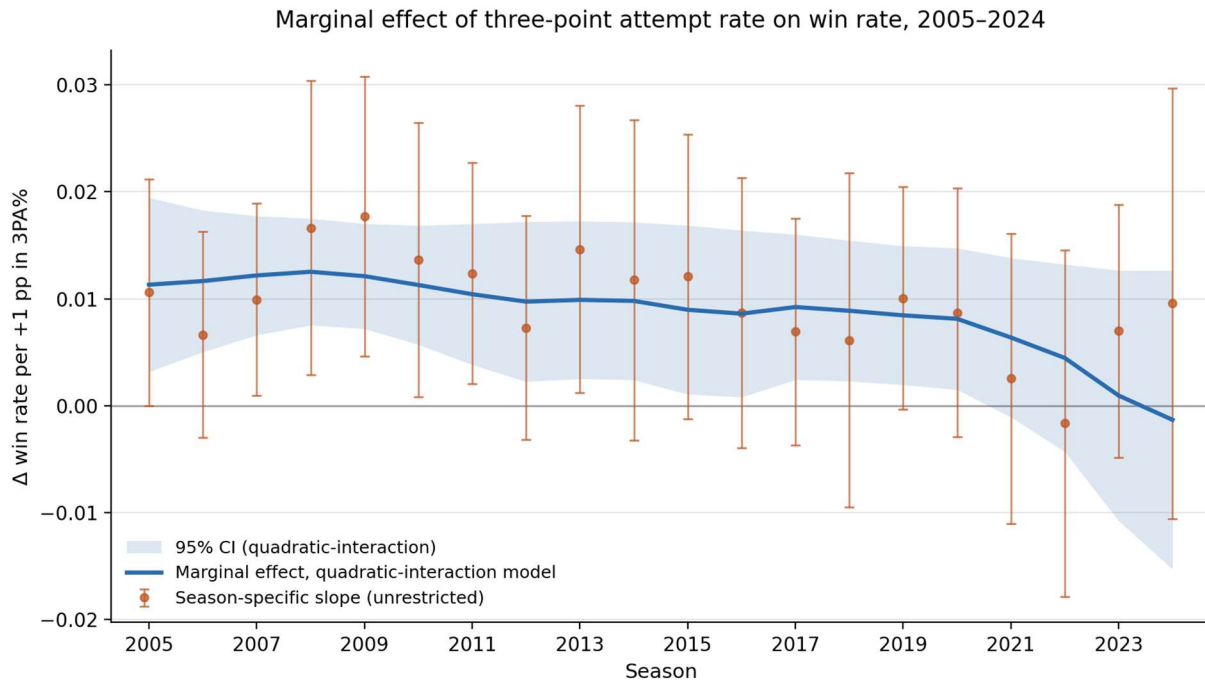


Figure 1. Marginal effect of the three-point attempt rate on win rate, 2005-2024. The solid line plots the marginal effect implied by column (4) of Table 3, evaluated at each season's league-mean attempt rate, with a 95 percent confidence band from the delta method. Points show the unrestricted season-specific slopes with 95 percent intervals. All quantities are expressed per one percentage point of 3PA%.

4.3 Robustness

The average return reported in column (2) is stable across a range of alternative choices. Excluding the two pandemic-affected seasons of 2020 and 2021 yields a coefficient of 0.994; restricting the sample to 2005 through 2019 yields 1.021; using the 35 raw team names in place of the 30 consolidated franchises yields 0.977; and replacing clustered standard errors with heteroskedasticity-robust ones leaves the point estimate unchanged and reduces the standard error from 0.237 to 0.148. Adding Pace as the sole additional regressor raises the coefficient to 1.114. The one specification in which the result weakens materially is the

restriction of the sample to 2015 through 2024, which yields 0.689 with a p-value of 0.103. That attenuation is not an anomaly but a further reflection of the pattern described above: in the second half of the sample there is both a smaller effect to detect and less variation with which to detect it.

5. Discussion

From an economic perspective, the results are less consistent with the classic account of information asymmetry followed by market correction than the initial framing of this paper anticipated, and more consistent with the slow diffusion of a technology whose value was never especially hidden. Under an informational account, the return to three-point volume should have been low before the innovation was recognized, high during the period in which early adopters held a temporary monopoly on strategic knowledge, and low again once the insight became common property. The estimates do not display that profile. The return is highest at the beginning of the sample, in the seasons immediately preceding the analytics era, and declines from there.

This reframing changes the question worth asking. If the marginal productivity of perimeter volume was already high in 2005, the puzzle is not why an advantage emerged in the mid-2010s but why nearly two decades were required to exhaust it. The most plausible explanation lies in adjustment costs rather than in information. A team's shot distribution is not a free parameter that a front office can reset between seasons. It is the product of roster composition, of coaching human capital, and of shooting skill accumulated by individual players over careers, each of which is a slow-moving stock. Recognizing that three-point attempts are undervalued is inexpensive; assembling the personnel to act on that recognition is not. Under this interpretation the Golden State Warriors and Houston Rockets of the mid-2010s were not the discoverers of a hidden inefficiency so much as the first organizations to complete the costly adjustment that the inefficiency rewarded.

The pattern of the 2020s admits a corresponding reading. As the entire league converged on high-volume perimeter shooting, teams adjusted their production functions toward the new standard, and the differential advantage available to any one of them contracted. In this saturated environment the attempt

rate is no longer an instrument of differentiation but a prerequisite for competition, and organizational effort shifts toward execution efficiency, roster optimization, and the identification of the next available margin. The estimates are consistent with this account, and the sharp decline in the cross-sectional dispersion of attempt rates provides independent descriptive support for it. It should be stated plainly, however, that the decline in the marginal effect is not statistically established, and the discussion in this section is offered as an interpretation of a suggestive pattern rather than as a conclusion the data compel.

5.1 The Rebirth of the Big Man

The league-wide saturation of three-point volume has had a heterogeneous impact across playing positions, most visibly in the evolution of the center. The observations that follow are illustrative rather than tested: this paper's data are team-season aggregates and cannot speak to position-level outcomes, and the discussion is included because it describes a mechanism consistent with the estimated pattern, not because the estimates establish it.

In the pre-analytics era, moderate league-wide attempt rates left room for traditional interior-focused big men to exert a steady influence on outcomes. This period represented the final chapters of dominance for centers such as Shaquille O'Neal, whose competitive value rested on physical dominance and interior scoring gravity rather than perimeter versatility. As perimeter volume accelerated between 2013 and 2018, these roles lost influence as teams prioritized pace and spacing.

The current period of strategic saturation has not resulted in the permanent marginalization of the position but in its redefinition. As defenses adapted to high-volume perimeter strategies, elite centers regained influence by combining interior gravity with perimeter efficiency and playmaking. Nikola Jokic, who won consecutive Most Valuable Player awards and an NBA championship during this period, is the clearest example: unlike the interior-bound centers of the early 2000s, he functions as a high-efficiency offensive hub capable of elite passing and perimeter shooting. This progression is consistent with the diffusion account offered above. Once every team has completed the adjustment toward perimeter volume,

the remaining margin lies not in shooting more three-pointers but in assembling personnel who can exploit the spacing that universal adoption creates.

5.2 Limitations

The most consequential limitation is the one already stated in the results. The paper's central hypothesis concerns time variation in the marginal effect, and the data do not permit that variation to be established. The point estimates decline and the descriptive evidence on convergence supports the same reading, but the formal tests, both parametric and unrestricted, fail to reject a constant effect. A longer panel, or one exploiting within-season variation, would be required to settle the question.

A second limitation concerns identification. Three-point attempt rate is chosen by teams rather than assigned, and the fixed-effects framework addresses only those confounders that are fixed within a franchise or common to a season. Time-varying, team-specific shocks remain. The clearest example is personnel: a team that acquires an exceptional shooter will simultaneously raise its attempt rate and improve its prospects, and the estimated coefficient will attribute to shot selection an effect that belongs in part to talent. The estimates should therefore be read as the association between attempt rate and winning that survives the removal of franchise and league-year effects, and only cautiously as the causal return to a strategic choice.

A third limitation follows from the decision to exclude contemporaneous efficiency measures. That decision is defended in Section 3.2 and is necessary if the estimate is to capture the full effect of shot selection, but it means the estimate is a total effect and that the paper cannot decompose it. Whether the return to perimeter volume operates through improved scoring efficiency, through the spacing it creates for other actions, or through its effect on opposing defensive structure cannot be determined within this framework, and separating these channels would require a design capable of distinguishing mediators from confounders rather than simply omitting both.

Fourth, the attempt rate and its square are severely collinear once fixed effects are removed, with variance inflation factors of 61 and 66, so their separate contributions cannot be distinguished. The joint

tests and the estimated derivative remain valid, but the individual coefficients on these two terms in column (4) of Table 3 carry little information.

Finally, the model shares two limitations with the original formulation. League-wide rule changes, such as the enforcement changes associated with freedom of movement, likely facilitated the rise in attempt rates and are absorbed by the season fixed effects in ways that complicate the isolation of pure strategy from its regulatory environment. And the reliance on team-season aggregates masks within-season volatility, injuries to key personnel, and mid-season roster changes. Future work using player-level tracking data could establish whether the apparent saturation reflects defensive adjustment, the exhaustion of available shooting talent, or physical limits on shooting accuracy at high volume.

6. Conclusion

This paper examines whether the return to three-point volume in the NBA has changed over the two decades spanning the 2005 to 2024 seasons. Using a balanced panel of 600 franchise-seasons with two-way fixed effects and standard errors clustered by franchise, it finds that shooting a larger share of field-goal attempts from beyond the arc than one's contemporaries is associated with a higher win rate. The estimated return is approximately 0.97 percentage points of win rate for each additional percentage point of attempt rate, or roughly 0.8 wins over a full season, and it is robust to the exclusion of pandemic-affected seasons, to sample restrictions, to the treatment of franchise relocations, and to the choice of variance estimator.

This finding stands in contrast to the result obtained under the specification originally employed, in which the estimated return was indistinguishable from zero. The difference is entirely attributable to the treatment of contemporaneous offensive and defensive efficiency. Those measures are realized outcomes of the same season's play and lie on the causal path from shot selection to winning, so conditioning on them removes the channel through which the strategy operates and drives the estimated return to zero by construction. The methodological point generalizes beyond this application: in settings where the outcome

is close to an accounting identity in a set of intermediate performance measures, including those measures as controls will mechanically eliminate the effect of any strategic variable that operates through them.

On the dynamic question that motivated the paper, the evidence is suggestive rather than conclusive. The estimated marginal effect declines monotonically across the sample and is statistically indistinguishable from zero in the final four seasons, and the cross-sectional dispersion of attempt rates has contracted sharply over the same period. Both patterns point toward a strategy that has been fully absorbed into league practice and no longer differentiates the teams that employ it. The formal tests, however, cannot reject the hypothesis that the return has been constant throughout, and the claim of saturation should be advanced only with that qualification attached.

What the results do call into question is the sequence assumed at the outset. The return to perimeter volume was not created by the analytics movement and then competed away; it was present, and largest, in the earliest seasons observed. The interesting feature of this episode is therefore not the emergence of a temporary informational rent but the length of time required for a competitive industry to exhaust a known and persistent advantage. Shot distribution is determined by roster composition, coaching human capital, and accumulated player skill, none of which adjusts quickly. Understood in this way, the small-ball transformation is less a story about the discovery of information than about the cost of acting on it, and it suggests that in professional sports, as in other competitive markets, the speed at which a superior technology diffuses is governed less by how difficult it is to recognize than by how difficult it is to implement.